
UNIFIED MECHANICS

Volume I

Foundations of Logical and Physical Action

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VOLUME I OF V

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A CORRIGIBLE EDITION. This text is corrigible. Earlier manifests and printings are retained as provenance, not as immutable authority. Corrections are recorded by version and supersession history; no scientific claim is frozen.

ABSTRACT

This volume reconstructs the corpus from its earliest conceptual commitments: distinction, identity, relation, self-description, retained structure, ordered traversal, physical record, and action. Exact algebra is separated from postulate, physical identification, and conjecture. The objective is not to infer modern physics from words alone, but to expose the minimal formal chain that the later volumes actually use and to mark every point at which additional mathematical or empirical structure enters.

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The scientific contract

1.1 What this series is

Unified Mechanics is reconstructed here as a *research programme*: a linked set of definitions, formal constructions, physical identifications, computations, and proposed tests. That description is deliberately narrower than a “theory of everything.” A mature physical theory must specify its state space, observables, dynamics, symmetries, dimensional constants, limiting regimes, and empirical likelihood. The present corpus contains parts of that structure, several exact finite constructions, and a number of ambitious bridges that are not yet complete. The purpose of this edition is to make the boundary between those categories visible at the point of use.

The series therefore asks three questions in sequence:

1. What follows exactly from the framework’s stated definitions and postulates?
2. Which parts reproduce or invoke established mathematics and physics?
3. Which additional bridge would be required before an internal construction became a physical explanation or prediction?

This order is more than editorial hygiene. It prevents a familiar error in foundational work: beginning with an evocative physical interpretation, performing correct algebra inside it, and then treating the correctness of the algebra as evidence for the interpretation. Algebra can certify a consequence of a model. It cannot, by itself, certify the model’s contact with nature.

Every boxed statement carries its epistemic type. Established result means a result from the external mathematical or physical literature, used under stated hypotheses. Internal result means a consequence of the framework’s own definitions or assumptions. Conditional identification marks a bridge from an internal object to a physical one. Conjecture is a precise open target. Exposure records a known defect, degeneracy, circularity, or missing derivation.

1.2 A living edition, not a frozen one

Earlier corpus releases used manifests and printing labels to identify a governing state. Those files remain valuable: they establish provenance, document when a statement entered the corpus, and allow an old calculation to be reproduced. They do not make a scientific conclusion immutable. In this edition, a claim may be corrected, narrowed, superseded, or withdrawn whenever its proof, physical bridge, or evidential basis changes.

The edition uses four records instead of a freeze:

Version history

identifies the text and date of each public release.

Supersession ledger

states which claim replaced which earlier wording, and why.

Dependency ledger

records which definitions and assumptions a result actually uses.

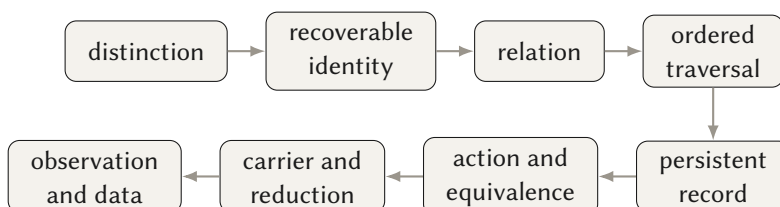
Evidence ledger

distinguishes derivation, computation, calibration, retrospective comparison, and prospective test.

This policy does not make the series unstable. It makes change explicit. A reader can reconstruct any prior state while treating the most recent scientifically graded statement as the current one.

1.3 The conceptual chronology

The corpus is most intelligible when read as a sequence of increasing physical commitment:



The first four terms are a proposed grammar of system description. A physical record adds dynamics and operational recoverability. An action adds variational or generative structure. A carrier adds a concrete mathematical representation. Observation finally connects the construction to instruments and data. Later volumes follow this same order rather than arranging material by the date or filename in which it happened to be written.

1.4 Authorship and external lineage

The framework and its terminology are Joseph Shields's. Its mathematical neighbourhood is not. Distinction has a particularly close historical contact with Spencer-Brown's calculus of indication (Spencer-Brown, 1969); self-reference and fixed-point arguments belong to a much broader logical tradition, for which Lawvere's categorical diagonal theorem is a useful landmark (Lawvere, 1969). Variational equivalence, quantum probability, causal reconstruction, lattice theory, gauge symmetry, and gravitational action have their own mature literatures. This edition cites those literatures where the corpus uses, parallels, or departs from them.

A citation is not an endorsement. Nor is it a claim that an older theorem anticipated Unified Mechanics. It identifies the exact point of contact, the hypotheses being borrowed, and the boundary beyond which the corpus is proposing something new.

Distinction, identity, and relation

2.1 The primitive vocabulary

The framework begins with three terms.

DEFINITION 2.1 — DISTINCTION

A *distinction* is whatever prevents complete identification of two admissible states or contents. Operationally, a distinction exists relative to a family of tests whenever at least one test yields different records for the two cases.

This operational clause is important. Without a class of tests, “different” is only a label. Let X be a state space and let $\mathcal{M}$ be a family of readout maps $m : X \rightarrow Y_m$. Define

$$x \sim_{\mathcal{M}} x' \iff m(x) = m(x') \text{ for every } m \in \mathcal{M}. \quad (2.1)$$

Then the recoverable objects are equivalence classes $[x]_{\mathcal{M}}$, not necessarily microscopic states. This supplies a precise form of the corpus’s statement that objecthood is retrospective: an object is what remains identifiable across the transformations and measurements admitted by the problem.

DEFINITION 2.2 — IDENTITY

Identity is recoverability of a distinction under a stated class of admissible transformations. If $\mathcal{T}$ is that class, identity requires that the relevant equivalence class remain discriminable along every $T \in \mathcal{T}$ for which persistence is claimed.

DEFINITION 2.3 — RELATION

A *relation* is an operation or structure through which distinct contents can be compared, coupled, transported, or transformed without being identified merely by participation in the relation.

These definitions are intentionally relational. They do not assert that the world is made only of distinctions, nor that every physical field is reducible to language. They specify the level at which the programme begins.

2.2 Relation to *Laws of Form*

Spencer-Brown begins with the injunction to draw a distinction and develops a calculus in which a mark can function as boundary, indication, crossing, and value (Spencer-Brown, 1969). Unified Mechanics adopts the priority of distinction but refuses that compression at the start. It keeps separate:

1. the act of drawing or producing a difference;
2. the boundary or relation thereby established;
3. the state indicated relative to that boundary;
4. the traversal across or through it;
5. the record by which the traversal remains recoverable.

This is a methodological difference, not a refutation of the calculus of indications. Spencer-Brown studies a compact algebra of forms. The present programme asks what physical machinery is required for a form to be propagated and read. Re-entry can make form behave dynamically; here, dynamics is kept explicit before form is compressed.

2.3 The seven operations

The inherited corpus gives two seven-term lists. The first describes the constitution of a system:

$$\text{distinguish} \prec \text{hold} \prec \text{close} \prec \text{answer} \prec \text{change} \prec \text{order} \prec \text{record}. \quad (2.2)$$

The second describes one traversal through an already articulated system:

$$\text{persist} \prec \text{distinguish} \prec \text{relate} \prec \text{propagate} \prec \text{resolve} \prec \text{return} \prec \text{close}. \quad (2.3)$$

Equation (2.2) is best read as a dependency chain. If those six precedence relations are adopted, the induced finite partial order has a unique linear extension: the displayed order. That is a correct but internal result. The substantive claim lies in the completeness of the seven operations, not in the order-theoretic proof after completeness has been assumed.

INTERNAL RESULT 2.1 — ORDERING OF THE PROPOSED GRAMMAR

Let $G = \{g_1, \dots, g_7\}$ and impose $g_i \prec g_{i+1}$ for $i = 1, \dots, 6$. The transitive closure is a total order and therefore has exactly one linear extension. This proves the ordering of the stipulated chain; it does not prove that every physical system must instantiate exactly these seven stages.

2.4 Systemhood as controlled unity

The first framework postulate is that a system may retain distinguishable internal contents without ceasing to be treated as one system. Mathematically, unity is then represented not by the absence of parts but by a closure condition: a chosen collection of states and operations remains within a specified domain. This is compatible with many familiar structures—a group closed under composition, an invariant subspace, a Markov process with a conserved state simplex, or a laboratory apparatus with a stable operational boundary. None is selected until further structure is supplied.

The second postulate is stronger: the first distinction does not occur in a pre-existing container; its retention supplies the first operational “where.” This is an ontological commitment. It may motivate an emergent account of space, but it is not yet geometry. A metric space needs a set, a distance or related structure, and appropriate axioms. A differentiable spacetime needs substantially more. Volume II therefore treats geometry as a later construction, not as a synonym for distinguishability.

EXPOSURE 2.1 — COMPLETENESS OF THE GRAMMAR

The corpus does not yet prove that the two seven-term lists are minimal, independent, or exhaustive. A formal completion requires (i) a syntax of admissible operations, (ii) equivalence rules, (iii) a demonstration that no operation is definable from the others, and (iv) a representation theorem showing that the intended systems instantiate the grammar.

Self-description and the golden fixed point

3.1 The update rule

The first exact numerical construction is generated by a two-component update. Write the state as $\mathbf{v}_n = (a_n, b_n)^\top$, where the first coordinate is the currently resolved share and the second is the held share. The inherited update is

$$\mathbf{v}_{n+1} = M\mathbf{v}_n, \quad M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}. \quad (3.1)$$

It adds the two available shares into the next resolved coordinate and carries one clean copy of the present resolved coordinate into reserve. The eigenvalues are

$$\lambda_{\pm} = \frac{1 \pm \sqrt{5}}{2}, \quad (3.2)$$

and the unique positive dominant eigenvalue is the golden ratio

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \varphi^2 = \varphi + 1. \quad (3.3)$$

INTERNAL RESULT 3.1 — SCALE-STABLE RATIO

For any initial vector with a nonzero component along the dominant eigenvector, repeated application of M , followed by normalization, converges to the positive eigenvector ratio φ . This is the Perron–Frobenius behaviour of the specified positive update matrix. The result selects φ only after M has been chosen.

The corpus sometimes describes this as the fixed point of self-description. Scientifically, the exact statement is narrower: φ is the stable ratio of the particular update in Equation (3.1). Calling that update the unique operation of physical self-description requires a separately proved uniqueness theorem over an explicitly defined admissible class.

3.2 The two-sided normalization

The principal numerical postulate of the corpus is a two-sided reading: a mixed relation is counted both from outward to held and from held to outward. With the inherited normalization this gives

$$r = \frac{1}{2\varphi} = \frac{\varphi - 1}{2} = \frac{\sqrt{5} - 1}{4}, \quad u = 1 - r = \frac{3 - \sqrt{5}}{4} + \frac{1}{2}. \quad (3.4)$$

Numerically,

$$r \approx 0.3090169944, \quad u \approx 0.6909830056. \quad (3.5)$$

The factor of two is not a consequence of Equation (3.3). It is the additional assumption A3. Keeping it visible prevents a genuine internal derivation from being misreported as parameter-free physics.

CONDITIONAL IDENTIFICATION 3.1 — RETAINED AND OUTWARD SHARES

If a physical process admits exactly two normalized channels and if its mixed relation is represented by the corpus's two-sided convention, then the channel coordinates are fixed to $r = (2\varphi)^{-1}$ and $u = 1 - r$. Evidence for the physical premise must be supplied separately for each application.

3.3 The three quadratic weights

Squaring the normalized partition yields

$$1 = (u + r)^2 = \underbrace{u^2}_{W_L} + \underbrace{2ur}_{W_B} + \underbrace{r^2}_{W_M}. \quad (3.6)$$

The exact and numerical values are

$$W_L = u^2 \approx 0.4774575141, \quad (3.7)$$

$$W_B = 2ur \approx 0.4270509831, \quad (3.8)$$

$$W_M = r^2 \approx 0.0954915028. \quad (3.9)$$

The three weights are exact once A3 is accepted. Their recurring appearance in later constructions is therefore a real consistency constraint: a later calculation cannot change their ratios without changing the foundation. Recurrence is not, however, independent evidence when the later formula was built from the same partition. The edition will distinguish *reuse of a prior* from *recovery from independent data*.

TABLE 3.1. The exact partition and the limits of its interpretation.

Term	Exact role	Physical status
$W_L = u^2$	diagonal outward contribution	internal algebra; “logical” is a label
$W_B = 2ur$	mixed cross term	internal algebra; boundary/interference meaning is conditional
$W_M = r^2$	diagonal retained contribution	internal algebra; matter or mass meaning is conditional

3.4 A minimal dependency graph

The numerical chain is short enough to display without rhetoric:

$$\begin{array}{c}
 \boxed{M} \implies \boxed{\varphi^2 = \varphi + 1}, \\
 \xrightarrow{\text{A3: two-sided normalization}} \boxed{r = (2\varphi)^{-1}, u = 1 - r} \implies \boxed{u^2, 2ur, r^2}.
 \end{array} \tag{3.10}$$

The only free conceptual step in the displayed numerical calculation is the choice and physical interpretation of M and A_3 . Everything to the right is exact algebra. That is the strongest scientifically defensible way to state the result.

Traversal, closure, and records

4.1 Traversal as an ordered map

A traversal is not simply an edge between two labelled vertices. It is a composable sequence

$$\tau = (x_0 \xrightarrow{f_1} x_1 \xrightarrow{f_2} \dots \xrightarrow{f_n} x_n) \quad (4.1)$$

together with the order in which its maps act. Two traversals can share endpoints and still differ through their intermediate states, their accumulated phase, their boundary data, or the records they leave. This elementary observation is the origin of the corpus's later insistence that bulk equations alone may be a coarse transcript of a physical action.

Let $\mathcal{O}$ denote the admissible future operations and $R : X \rightarrow \mathcal{Y}$ a readout. A distinction between x and x' is persistent over $\mathcal{O}$ if

$$\forall O \in \mathcal{O}_{\text{relevant}}, \quad R(Ox) \neq R(Ox'). \quad (4.2)$$

The definition is relative to a readout and a future-operation class. In noisy physics one normally replaces the sharp inequality by a statistical discriminability criterion. The binary record of the inherited corpus is therefore an ideal limiting case.

DEFINITION 4.1 — RECORD

A record is a physical state whose distinction from the relevant alternatives remains recoverable under a declared family of subsequent operations and readouts.

This definition accommodates a detector latch, a memory bit, a track in a chamber, a durable macroscopic correlation, or a stored data product. It also reveals why a record cannot be inferred from a label: persistence is a dynamical property.

4.2 Closure and return

The corpus distinguishes *return* from *closure*. A path returns when its projected endpoint agrees with its starting point. It closes as a physical record only if all state variables relevant to identity—including order, phase, boundary data, or memory—satisfy the stated closure criterion. A periodic coordinate can therefore hide a nonperiodic history.

If $\pi : \tilde{X} \rightarrow X$ forgets a history coordinate, then it is possible that

$$\pi(\tilde{x}_n) = \pi(\tilde{x}_0) \quad \text{while} \quad \tilde{x}_n \neq \tilde{x}_0. \quad (4.3)$$

This is ordinary mathematics of a lift or covering representation. Volume III asks when such a lifted coordinate is physically measurable and whether it can be made covariant.

4.3 The proposed crossing quantum

The inherited register associates one successful primitive crossing with a retained factor r , and a minimal three-crossing closed history with

$$r^3 = \frac{1}{8\varphi^3} = \frac{\sqrt{5}-2}{8} \approx 0.0295084972. \quad (4.4)$$

The identity is exact. The quantisation claim is not yet a theorem. The binary definition of a record tells us when the chosen readout treats a distinction as recovered; it does not entail that nature assigns the numerical cost r to every successful crossing.

EXPOSURE 4.1 — MISSING COST FUNCTIONAL

To derive Equation (4.4) as physics, the programme must define a cost, probability, amplitude, or conserved current for a primitive crossing; show why its value is exactly r ; and demonstrate why three crossings, rather than any other admissible history, govern the target observable. Until then, r^3 is an exact internal coordinate and a candidate parameter-free threshold.

4.4 Retention and the arrow

The corpus's retention principle says that a continued self-description joins a new resolution to the existing record rather than replacing that record. This is a coherent identity criterion: if the state at step $n+1$ contains no recoverable relation to the state at step n , the phrase “the same system continued” has lost its operational basis.

The principle does not prove a universal thermodynamic arrow. Microscopic reversible dynamics, open-system entropy production, cosmological time orientation, and the psychological arrow are different questions. Landauer’s bound connects logically irreversible erasure to thermodynamic cost under explicit physical assumptions (Landauer, 1961); Bennett showed how reversible computation clarifies that connection (Bennett, 1982). Those results support careful separation of logical and thermodynamic irreversibility, not their automatic identification.

CONDITIONAL IDENTIFICATION 4.1 — ORDER AS PHYSICAL TIME

If persistent record formation is monotone along a single physical parameter, that parameter supplies an operational time orientation for the recording subsystem. Establishing a unique global time for an arbitrary spacetime requires additional causal and geometric hypotheses.

From evolution to action

5.1 The action principle and its quotient

For fields ϕ^a on a domain M , an action is a functional

$$S[\phi] = \int_M L(x, \phi, \partial\phi, \dots) d^d x. \quad (5.1)$$

Its Euler–Lagrange expressions are

$$E_a(L) = \sum_{|I| \geq 0} (-D)_I \left(\frac{\partial L}{\partial (D_I \phi^a)} \right). \quad (5.2)$$

Two Lagrangian densities can give the same bulk equations. Locally, adding a total divergence,

$$L' = L + D_\mu B^\mu, \quad (5.3)$$

does not change Equation (5.2) for compactly supported variations. This motivates an equivalent-action relation

$$L \sim_{\text{EA}} L' \iff E(L) = E(L'). \quad (5.4)$$

Noether’s theorem relates differentiable variational symmetries to conserved currents (Noether, 1918). The variational bicomplex gives a systematic language for Euler–Lagrange forms, horizontal divergences, and the inverse problem (Anderson, 1992; Olver, 1993). The important scientific point is negative as well as positive: not every differential equation is Euler–Lagrange in its given variables. The Helmholtz conditions test whether a source form is locally variational.

ESTABLISHED RESULT 5.1 — THE INVERSE-VARIATIONAL GATE

A system of differential equations is locally Euler–Lagrange only when its linearisation satisfies the appropriate formal self-adjointness/Helmholtz conditions, possibly after a nondegenerate multiplier is introduced. Dissipative equations such as the heat equation do not generally arise from an ordinary local action of the same field alone.

5.2 Why bulk equivalence can be too coarse

Equation (5.4) forgets boundary data. In general relativity the Einstein–Hilbert action contains second derivatives of the metric; a well-posed Dirichlet variational principle on a manifold with boundary requires the Gibbons–Hawking–York term (York, 1972; Gibbons and Hawking, 1977). Two expressions may therefore agree in their bulk Euler–Lagrange equations yet differ in boundary variation, conserved charge, semiclassical action, or thermodynamic interpretation.

This gives a rigorous core to the corpus’s “transcription” critique. A transcript is adequate for a target question only if it preserves every distinction on which the answer depends.

INTERNAL RESULT 5.1 — SUFFICIENCY OF A COMPRESSION

Let $C : X \rightarrow Z$ be a compression and $Q : X \rightarrow Y$ a target quantity. Then Q is exactly recoverable from C if and only if Q is constant on every fibre of C , equivalently if and only if there exists $\tilde{Q} : Z \rightarrow Y$ such that

$$Q = \tilde{Q} \circ C. \quad (5.5)$$

This is the precise content of the corpus’s field-sufficiency rule.

If two physical preparations collapse to the same compressed description but later produce different target values, the compression is harmful for that target. If no admissible future operation separates them, the collapse is harmless relative to the declared experiment.

5.3 The principle of logical action

The corpus proposes a refinement of bulk action equivalence. In addition to the Euler–Lagrange operator, a complete physical transcript may need to retain

$$\Lambda[L]equiv(E(L), \Theta_L, \mathcal{B}_L, \mathcal{O}_L, \mathcal{R}_L), \quad (5.6)$$

where Θ_L is the symplectic-potential/boundary variation, $\mathcal{B}_L$ denotes boundary conditions and boundary terms, $\mathcal{O}_L$ records execution or causal ordering when relevant, and $\mathcal{R}_L$ denotes the readout relations the action is intended to preserve. The exact list is programme-dependent; Equation (5.6) makes that dependence inspectable.

DEFINITION 5.1 — LOGICAL-ACTION EQUIVALENCE

For a declared target class $\mathcal{Q}$, write $L \sim_{\text{LA}} L'$ when their complete traces agree on every component required to recover all $Q \in \mathcal{Q}$. Logical-action equivalence is therefore finer than equality of bulk equations whenever boundary, order, or record data affect the target.

This definition avoids claiming that one universal maximal transcript has already been found. The scientific task is to state the target first and prove sufficiency by factorisation as in Equation (5.5).

RESEARCH TARGET 5.1 — MINIMAL PHYSICAL TRANSCRIPT

Classify, for major field theories, the smallest extension of the Euler–Lagrange quotient that preserves boundary charges, causal composition, measurement couplings, and empirically distinguishable histories. A successful classification would turn the corpus’s Principle of Logical Action from a methodological rule into a representation theorem.

The quantum interface

6.1 What ignorance does not prove

Suppose two alternatives remain compatible with the available record. One may represent that situation by a classical probability distribution, a coherent quantum state, a set of possible models, or a more general convex state. The absence of a distinguishing event does not select among those structures. Accordingly, the inherited statement “superposition is forced” is too strong if based only on nonselection.

A quantum reconstruction needs operational assumptions that distinguish coherent composition from classical mixing. Examples include continuous reversible transformations, local tomography, purification, or interference structure; modern reconstructions make such assumptions explicit (Chiribella et al., 2011). The framework may seek to motivate them through traversal and retention, but it cannot omit them.

EXPOSURE 6.1 — SUPERPOSITION IS NOT A CONSEQUENCE OF IGNORANCE

The proposition “no event selected an alternative” is compatible with both a density matrix containing coherence and a classical mixture diagonal in the same basis. An empirical or operational axiom sensitive to interference is required to distinguish them.

6.2 When complex phase follows

There is nevertheless a clean conditional result behind the corpus’s phase argument. Let an additive action parameter $S \in \mathbb{R}$ label sequential composition, and let $\chi : \mathbb{R} \rightarrow U(1)$ be a continuous homomorphism:

$$\chi(S_1 + S_2) = \chi(S_1)\chi(S_2), \quad |\chi(S)| = 1. \quad (6.1)$$

Then

$$\chi(S) = e^{i\kappa S} \quad (6.2)$$

for some real constant κ . With $\kappa = 1/\hbar$, Equation (6.2) is the familiar quantum phase assigned to an action history, central to Feynman’s spacetime formulation

(Feynman, 1948).

INTERNAL RESULT 6.1 — CONTINUOUS PHASE CHARACTER

Complex exponential phase is the unique continuous one-dimensional unitary representation of additive real action, up to the scale κ . The premises already contain the unit circle and reversible multiplicative composition; the theorem does not derive those premises or the value of $\hbar$.

6.3 Probability: the precise Gleason route

Born introduced the probabilistic interpretation of quantum amplitudes in scattering theory (Born, 1926). Gleason later proved that, on a real or complex Hilbert space of dimension at least three, a countably additive nonnegative measure on the lattice of closed subspaces has the trace form

$$p(P) = \text{Tr}(\rho P) \quad (6.3)$$

for a positive trace-class operator ρ (Gleason, 1957). Normalization fixes $\text{Tr } \rho = 1$. Generalized-measurement versions obtain the trace rule for effects, including qubit settings under the corresponding assumptions (Busch, 2003; Caves et al., 2004).

ESTABLISHED RESULT 6.1 — NONCONTEXTUAL QUANTUM PROBABILITY

For the measurement domains covered by Gleason-type theorems, additive probabilities assigned noncontextually to the same projector or effect have the Born trace form. The theorem presupposes Hilbert-space event structure and noncontextual additivity; it is not a derivation of Hilbert space from the seven-operation grammar.

The corpus's phrase "the same physical effect carries the same probability in every complete context" is a readable statement of measurement noncontextuality. The final edition retains that intuition but places the theorem's dimension and event-space hypotheses beside it.

6.4 Symmetry, unitarity, and the generator

Wigner's theorem states that a bijection of rays preserving transition probabilities is induced by a unitary or antiunitary transformation (Wigner, 1959). A continuous one-parameter family connected to the identity is represented unitarily. Stone's theorem then states that every strongly continuous one-parameter unitary group $U(t)$ has a unique self-adjoint generator H such

that

$$U(t) = e^{-itH} \quad (6.4)$$

in mathematical units (Stone, 1932). Restoring $\hbar$ and identifying H with energy gives

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle. \quad (6.5)$$

ESTABLISHED RESULT 6.2 — THE WIGNER–STONE CHAIN

Transition-probability preservation plus bijectivity gives a unitary or antiunitary implementation; strong continuity of a one-parameter unitary group gives a self-adjoint generator. The physical identification of the parameter with time and the generator with energy is additional structure.

6.5 Measurement and record formation

An ideal premeasurement correlates a system basis with apparatus pointer states,

$$\left(\sum_i c_i |s_i\rangle \right) |M_0\rangle \longrightarrow \sum_i c_i |s_i\rangle |M_i\rangle. \quad (6.6)$$

This unitary interaction does not by itself select one term. Environmental decoherence can suppress interference between robust pointer records in reduced states, explaining the practical stability of classical alternatives (Zurek, 2003); general instruments describe conditioned state updates and outcome statistics (Ozawa, 1984). Neither fact licenses collapsing premeasurement, decoherence, conditioning, and human awareness into one unnamed event.

The framework’s observer definition is useful precisely when made mechanical: a prepared ready state, a coupling that defines the channel, propagation, amplification, resolution, and a persistent record. Volume III develops that chain and the information lost at each stage.

6.6 The honest quantum ledger

TABLE 6.1. What the foundational chain does and does not establish.

Statement	Grade	Required hypotheses or missing bridge
Continuous additive action has exponential phase	internal conditional result	representation into $U(1)$, continuity, reversible composition
Born trace rule	established external result	Hilbert effects/projectors, additivity, noncontextuality, dimensional conditions
Closed continuous evolution has a self-adjoint generator	established external result	ray symmetry, bijectivity, strong continuity, one-parameter group
The seven-operation grammar uniquely yields quantum theory	open	operational reconstruction including composition, interference, tomography, and state-space axioms
Definite outcomes follow from record persistence	open	interpretive measurement model beyond unitary premeasurement and decoherence

Information, compression, and bounded recovery

7.1 Records are outputs of channels

Let a world state x generate a record y through a measurement channel M :

$$y = M(x) + \epsilon, \tag{7.1}$$

where ϵ represents declared noise or model discrepancy. Any distinction in the exact kernel of a linear M , or more generally in a fibre of a nonlinear M , is absent from the record. No decoder using y alone can determine which member of that fibre occurred. This is a direct consequence of factorisation, and in probabilistic settings it is expressed by data-processing inequalities in information theory (Shannon, 1948; Cover and Thomas, 2006).

ESTABLISHED PRINCIPLE 7.1 — NO RECOVERY WITHOUT TRANSMITTED INFORMATION OR A PRIOR

If two admissible states induce the same distribution of records, no estimator based only on those records can distinguish the states. Recovery can become possible only by changing the channel, adding data, or restricting the admissible state ensemble.

This principle disciplines the corpus’s observational language. A telescope, detector, simulation catalogue, and human classification do not reveal “the object itself.” Each yields a record through a mechanism with a passband, noise model, point-spread function, selection function, and null space.

7.2 Recovery under a prior

Compressed sensing demonstrates how an underdetermined linear problem can become well posed when the unknown is sparse in a known representation and the measurement operator satisfies suitable conditions (Donoho, 2006;

Candès et al., 2006). Schematically,

$$\hat{x} = \arg \min_z \|z\|_1 \quad \text{subject to} \quad \|Mz - y\|_2 \leq \delta. \quad (7.2)$$

The prior does not restore information destroyed for an unrestricted state. It restricts the candidate ensemble so that the surviving measurements identify one member of that smaller class.

Unified Mechanics proposes its channel grammar and fixed weights as such a structural prior. That is a coherent programme, but it is not automatically compressed sensing. For each application the edition will ask:

1. What is the measurement map M ?
2. What class of states is restricted by the prior?
3. What theorem makes the inverse unique or stable on that class?
4. How many adjustable parameters enter the map to the observable?
5. Was the map fixed before inspecting the evaluation data?

7.3 The zero-knob rule

A parameter-free map can make a sharp forecast, but “zero fitted parameters” is not sufficient for evidential force. A mapping may contain hidden discrete choices, selected normalisations, flexible data cuts, or an observable chosen after inspection. Accordingly, the governing rule is:

DEFINITION 7.1 — ADMISSIBLE FORECAST

A forecast is admissible as parameter-free only when the source-to-observable map, all discrete choices, normalization conventions, data cuts, loss function, and success criterion were fixed before the evaluation data were inspected. Otherwise the result is a retrospective comparison or a postdiction.

The rule applies equally to positive and negative outcomes. A non-identifiable test should be rejected before execution when possible. A test whose failure is reinterpreted only afterward does not become informative through narrative repair.

7.4 Harmless and harmful quotients

Let $C : X \rightarrow Z$ collapse states into equivalence classes. For a family of future quantities $\mathcal{Q}$, the collapse is harmless if every $Q \in \mathcal{Q}$ factors through C . It is harmful if some admissible target separates two states in one fibre:

$$\exists x, x', Q : \text{quad} C(x) = C(x') \quad \text{and} \quad Q(x) \neq Q(x'). \quad (7.3)$$

This criterion unifies several themes in the corpus: boundary terms erased by the bulk Euler–Lagrange map, phases erased by an incoherent mixture, traversal depth erased by a periodic projection, and physical distinctions erased by a sensor.

INTERNAL RESULT 7.1 — TARGET-RELATIVE KERNEL STABILITY

A kernel is harmless relative to a future dynamics F_t and readout R when $C(x) = C(x')$ implies $R(F_t x) = R(F_t x')$ for every relevant t . It is harmful when the implication fails. This is a target-relative criterion, not an absolute property of the compression.

The foundational register

8.1 Minimal commitments

The reconstructed foundation can be stated without importing later particle, gravitational, or cosmological claims.

TABLE 8.1. Foundational commitments and their living-edition status.

ID	Name	Status and content
A1	systemhood	framework postulate: internal difference is compatible with operational unity
A2	ground	ontological postulate: retained distinction supplies the first operational locus
A3	two-sided reading	numerical postulate fixing the factor of two in $r = (2\varphi)^{-1}$
A4	saturation	open physical postulate requiring a defined capacity and threshold
A5	conserved read-out	physical postulate requiring a measure or current in each realisation
A6	admissible reduction	operator-algebra definition: unital, idempotent, completely positive, bimodular, and co-variant

The first three generate the conceptual and numerical foundation. A4 and A5 become relevant only with dynamics. A6 belongs to the carrier and readable-algebra construction of Volume II.

8.2 Core definitions

TABLE 8.2. Core vocabulary, condensed from the inherited register.

ID	Term	Operational definition
D1	distinction	a difference recoverable by at least one declared test
D2	identity	recoverability of a distinction under admissible change
D3	relation	comparison, coupling, or transformation without automatic identification
D4	seven doings	distinguish, hold, close, answer, change, order, record
D5	traversal stages	persist, distinguish, relate, propagate, resolve, return, close
D6	object	a stable equivalence class under declared transformations and readouts
D7	update	$(a, b) \mapsto (a + b, a)$, the chosen two-channel recurrence
D8	channels	outward share u and retained share r , with $u + r = 1$
D9	weights	$W_L = u^2$, $W_B = 2ur$, $W_M = r^2$
D10	crossing	a primitive enacted traversal; its proposed price is not yet derived
D11	record	a physically retained distinction recoverable under relevant future operations
D12	observer	a system that couples to, resolves, amplifies, and persistently retains another distinction
D13	carrier	the mathematical structure on which records are represented; self-duality is a framework requirement
D14	primitive crossing	a smallest nonzero admissible carrier displacement, once a carrier and norm are specified
D15	cluster projection	admissible reduction onto a readable subalgebra
D16	decay generator	a positive operator whose kernel is the readable algebra; minimal candidate $I - C$
D17	readable algebra	the fixed/range algebra retained by the reduction

ID	Term	Operational definition
D18	geometry projectors	proposed mutually orthogonal external, temporal, and closure sectors

8.3 What Volume I establishes

INTERNAL SYNTHESIS

The specified update has the golden ratio as its stable positive eigenvalue. With the additional two-sided normalization A_3 it fixes r , u , and the three quadratic weights exactly. The framework also supplies a target-relative language of identity, records, compression, and traversal, plus a refined action-equivalence programme that keeps boundary and order data when a target depends on them.

FOUNDATIONAL OBLIGATIONS CARRIED FORWARD

The series has not derived the completeness of the seven-operation grammar, the physical value of a crossing, the uniqueness of a global time mode, Hilbert-space quantum theory from the grammar alone, or a universal minimal logical-action transcript. These are explicit research obligations, not conclusions hidden behind later notation.

8.4 Forward link

Volume II asks whether the abstract carrier requirements—finite primitive crossings, self-duality, an admissible reduction, and a readable algebra—select a concrete exceptional structure. That step introduces lattice integrality, evenness, rank, root systems, Coxeter geometry, and field-theoretic identifications. None of those properties is smuggled into the foundation retroactively; each appears where its new assumption or theorem is first required.

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